

1. (判断题) 判断以下关于线性映射的说法是否正确:

- (1) 线性映射 A 可逆, 则其行列式不为零;
- (2) 线性映射 A, B 满足 $\text{rank}(A) = \text{rank}(B)$, 则 $\text{rank}(A^2) = \text{rank}(B^2)$;
- (3) $(A+B)^2 = A^2 + 2AB + B^2$;
- (4) $\text{rank}(A) + \text{rank}(B) \geq \text{rank}(A+B)$;
- (5) 设 A 矩阵在一组基下矩阵表示为 $\mathbf{A}$, B 矩阵在这组基下矩阵表示为 $\mathbf{B}$, 则线性映射 AB 在这组基下矩阵表示为 $\mathbf{AB}$ 。

(1) $\checkmark$ (2) $\times$ (3) $\times$ (4) $\checkmark$ (5) $\checkmark$

6. (综合题) 考虑线性空间 $V = \{A \in M_{2 \times 2}(\mathbb{K}) \mid \text{tr}(A) = 0\}$ 。给定矩阵 $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, 定义线性变换 $\Phi_P(X) = P^{-1}XP$ 。

- (1) 求证: $\{E_{11} - E_{22}, E_{12}, E_{21}\}$ 是 V 的一组基;
- (2) 求出线性变换 Φ_P 在这组基下的矩阵;
- (3) 求出线性变换 Φ_P 的特征值与特征向量;

6. (2) $P^{-1} = P$.

$$P^{-1} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = -(E_{11} - E_{22})$$

$$P^{-1} E_{12} P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = E_{21}$$

$$P^{-1} E_{21} P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = E_{12}$$

$$\therefore \text{对应矩阵为 } \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$(3) |\lambda I - A| = \begin{vmatrix} \lambda + 1 & & \\ & \lambda & -1 \\ & -1 & \lambda \end{vmatrix} = (\lambda + 1) \cdot (\lambda^2 - 1)$$

$\therefore \lambda = -1$ (双根), 1 ; 记基为 $\varepsilon_1, \varepsilon_2, \varepsilon_3$.

① $\lambda = -1$ 时, $\begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ 秩为 1, $\eta_1 = \varepsilon_1, \eta_2 = \varepsilon_2 - \varepsilon_3$
特征向量为 $k_1 \eta_1 + k_2 \eta_2$

② $\lambda = 1$ 时, $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ $\eta_3 = \varepsilon_2 + \varepsilon_3$; 特征向量为 $k \eta_3$

(4) 推广: 若 $\Phi_U(X) = U^{-1}XU$, 且可逆矩阵 U 有两个特征值 2, 4, 求出线性变换 Φ_U 的特征值。

7. (证明题) 设 A 是正定矩阵, B 是实对称矩阵。求证: 存在可逆矩阵 C , 使得 $C^T A C$ 与 $C^T B C$ 同时为对角矩阵。

(4) 由于 U 有两个特征值, $\therefore U$ 可相似对角化

$$U = P^{-1} \cdot \text{diag. } \{2, 4\} \cdot P$$

X 变换后与原矩阵相似; 由于秩为 2, 则其特征子空间维数均为 1

$$\begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}^{-1} = \frac{1}{8} \cdot \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$U^{-1} \cdot E_{12} \cdot U = \frac{1}{2} E_{12}$$

$$U^{-1} E_{21} U = 2 E_{21}$$

$\therefore$ 特征值为 $1, 2, \frac{1}{2}$ 相似变换不改变特征值

7. $\therefore A$ 正定, $\therefore A$ 标准形矩阵为 I

$$\therefore P^T A P = I \quad P \text{ 可逆}$$

$\therefore B$ 为实对称矩阵, $\therefore P^T B P$ 亦为实对称矩阵

$$\therefore T(P^T B P)T = D$$

$$\text{令 } C = PT$$

$$\therefore C^T A C = T^T (P^T A P) T = T^T I = I$$

$$C^T B C = D$$

1. $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}$ 行列式为范德蒙行列式, 一定可逆

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 4 & 0 & 1 & 0 \\ 1 & 3 & 9 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{\substack{\textcircled{2}-\textcircled{1} \\ \textcircled{3}-\textcircled{1}}} \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 2 & 8 & -1 & 0 & 1 \end{pmatrix} \xrightarrow{\textcircled{3}-\textcircled{2} \times 2} \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -2 & 1 \end{pmatrix}$$

$$\xrightarrow{\textcircled{3} \times \frac{1}{2}} \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & -1 & \frac{1}{2} \end{pmatrix} \xrightarrow{\substack{\textcircled{1}-\textcircled{3} \\ \textcircled{2}-\textcircled{3} \times 3}} \begin{pmatrix} 1 & 1 & 0 & \frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{5}{2} & 4 & -\frac{3}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -1 & \frac{1}{2} \end{pmatrix}$$

$$\xrightarrow{\textcircled{1}-\textcircled{2}} \begin{pmatrix} 1 & 0 & 0 & 3 & -3 & 1 \\ 0 & 1 & 0 & -\frac{5}{2} & 4 & -\frac{3}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -1 & \frac{1}{2} \end{pmatrix}$$

$$\therefore A^{-1} = \begin{pmatrix} 3 & -3 & 1 \\ -\frac{5}{2} & 4 & -\frac{3}{2} \\ \frac{1}{2} & -1 & \frac{1}{2} \end{pmatrix}$$

2. A 为实对称矩阵

$$|\lambda I - A| = \begin{vmatrix} \lambda-1 & -2 & -2 \\ -2 & \lambda-1 & 2 \\ -2 & 2 & \lambda-1 \end{vmatrix} = (\lambda-1) \begin{vmatrix} \lambda-1 & 2 \\ 2 & \lambda-1 \end{vmatrix} + 2 \begin{vmatrix} -2 & -2 \\ 2 & \lambda-1 \end{vmatrix} - 2 \begin{vmatrix} -2 & -2 \\ \lambda-1 & 2 \end{vmatrix}$$

$$= (\lambda-1) \cdot (\lambda^2 - 2\lambda + 1 - 4) + 4 \cdot (2 - 2\lambda + 4)$$

$$= \lambda^3 - 2\lambda^2 - 3\lambda - \lambda^2 + 2\lambda + 3 + 24 - 8\lambda$$

$$= \lambda^3 - 3\lambda^2 - 9\lambda + 27$$

$$= \lambda^2(\lambda-3) - 9(\lambda-3) = (\lambda^2-9)(\lambda-3) = (\lambda+3)(\lambda-3)^2$$

$$\textcircled{1} \lambda=3 \text{ 时, (三重)} \begin{bmatrix} 2 & -2 & -2 \\ -2 & 2 & 2 \\ -2 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \therefore \begin{matrix} x_1 = x_2 + x_3 \\ a_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad a_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \end{matrix}$$

$$\beta_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \beta_2 = a_2 - \frac{(a_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

$$\textcircled{2} \lambda=-3 \text{ 时, } \begin{bmatrix} -4 & -2 & -2 \\ -2 & -4 & 2 \\ -2 & 2 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 \\ 0 & 6 & -6 \\ 0 & -6 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad a_3 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\therefore T = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{1}{\sqrt{6}} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{1}{\sqrt{6}} & -\frac{\sqrt{3}}{3} \\ 0 & \frac{2}{\sqrt{6}} & -\frac{\sqrt{3}}{3} \end{bmatrix} \quad T^{-1}AT = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$3, \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow[\textcircled{1}-\textcircled{2}]{\textcircled{1}-\textcircled{2}} \begin{pmatrix} -4 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow[\textcircled{1}-\textcircled{3}]{\textcircled{1}-\textcircled{3}} \begin{pmatrix} -4 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \xrightarrow[\textcircled{2}+\textcircled{3}]{\textcircled{2}+\textcircled{3}} \begin{pmatrix} -4 & 0 & 0 \\ 0 & 4 & 2 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix}$$

$$\xrightarrow[\textcircled{2} \times \frac{1}{2}]{\textcircled{2} \times \frac{1}{2}} \begin{pmatrix} -4 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & \frac{1}{2} & 0 \\ -1 & \frac{1}{2} & 1 \end{pmatrix} \xrightarrow[\textcircled{3}-\textcircled{2}]{\textcircled{3}-\textcircled{2}} \begin{pmatrix} -4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \\ -1 & \frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$\therefore$ 正惯性系数为 1, 负惯性系数为 2.

$$4, \dim(V_1 \cap V_2) = \dim(V_1) + \dim(V_2) - \dim(V_1 + V_2)$$

$$V_1 + V_2 = \langle \alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3 \rangle$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 1 \\ -1 & 0 & 1 & 1 & -1 & -1 \\ 2 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 1 & 2 & 0 \end{bmatrix} \xrightarrow[\textcircled{3}-2\times\textcircled{1}]{\textcircled{1}-\textcircled{1}, \textcircled{2}+\textcircled{1}} \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & -1 \\ 0 & 1 & 1 & 1 & 1 & -1 \end{bmatrix} \xrightarrow{\textcircled{3}-\textcircled{2}} \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\therefore \dim(V_1 + V_2) = 4$$

且 $\alpha_1, \alpha_2, \alpha_3, \beta_2$ 构成 $V_1 + V_2$ 的一组基

$$\therefore \dim(V_1 \cap V_2) = 2.$$

$$\text{可以看出 } \alpha_1 - \alpha_2 + 2\alpha_3 = \beta_1$$

$$\therefore \beta_1 \in V_1 \cap V_2$$

$$\text{设 } \beta_3 = k_1 \alpha_1 + k_2 \alpha_2 + k_3 \alpha_3 + k_4 \beta_2$$

$$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ -1 & 0 & 1 & -1 & -1 \\ 2 & 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\therefore \beta_3 = \alpha_1 + \alpha_2 - \alpha_3 - \beta_2$$

$$\therefore \beta_2 + \beta_3 = \alpha_1 + \alpha_2 - \alpha_3, \quad \beta_2 + \beta_3 \in V_1 \cap V_2$$

而 β_1 与 $\beta_2 + \beta_3$ 线性无关
 $\therefore \beta_1$ 与 $\beta_2 + \beta_3$ 是一组基

$$\begin{bmatrix} 1 & -1 & 2 \\ 1 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \end{bmatrix}$$

rank = 2. 满秩.

$$5. (\delta_1, \delta_2, \delta_3) = (\varepsilon_1, \varepsilon_2, \varepsilon_3) \cdot P$$

$$(\eta_1, \eta_2, \eta_3) = (\varepsilon_1, \varepsilon_2, \varepsilon_3) \cdot Q$$

$$1) \vec{A}(\varepsilon_1, \varepsilon_2, \varepsilon_3) = (\varepsilon_1, \varepsilon_2, \varepsilon_3) \cdot A$$

$$\begin{cases} \eta_1 = 2\varepsilon_1 - \varepsilon_2 - \varepsilon_3 \\ \eta_2 = -2\varepsilon_1 + 2\varepsilon_2 - \varepsilon_3 \\ \eta_3 = \varepsilon_1 - \varepsilon_2 + \varepsilon_3 \end{cases}$$

$$\vec{A}(\eta_1, \eta_2, \eta_3) = \vec{A}(2\varepsilon_1 - \varepsilon_2 - \varepsilon_3, -2\varepsilon_1 + 2\varepsilon_2 - \varepsilon_3, \varepsilon_1 - \varepsilon_2 + \varepsilon_3)$$

$$= (\varepsilon_1, \varepsilon_2, \varepsilon_3) \begin{bmatrix} 4 & -6 & 1 \\ -2 & 6 & -1 \\ -2 & -3 & 2 \end{bmatrix} = (\eta_1, \eta_2, \eta_3) \cdot Q^{-1} \begin{bmatrix} 4 & -6 & 1 \\ -2 & 6 & -1 \\ -2 & -3 & 2 \end{bmatrix}$$

$$|Q| = 2 \cdot x \cdot | + | x \cdot \begin{vmatrix} -2 & 1 \\ -1 & 1 \end{vmatrix} - 1 \cdot \begin{vmatrix} -2 & -1 \\ 2 & -1 \end{vmatrix}$$

$$= 2 + (-1) = 1$$

$$Q^{-1} = Q^* = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 1 \\ 3 & 4 & 2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 1 \\ 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} 4 & -6 & 1 \\ -2 & 6 & -1 \\ -2 & -3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$

$\therefore$ 在 η_1, η_2, η_3 下的矩阵为

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$

$$(2) (\delta_1, \delta_2, \delta_3) = (\varepsilon_1, \varepsilon_2, \varepsilon_3) \cdot P = (\eta_1, \eta_2, \eta_3) Q^{-1} \cdot P$$

$$Q^{-1} \cdot P = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 1 \\ 3 & 4 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & -1 \\ -3 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 & -1 \\ 5 & 6 & -1 \\ 5 & 8 & 0 \end{bmatrix}$$

6. (1) X 可逆等价于 $|X| \neq 0$

若 C 不可逆, 经过初等行变换后, X 存在 0 行, $|X| = 0$, $\therefore C$ 可逆

同理 B 一定要可逆.

$$\begin{pmatrix} A & B \\ C & 0 \end{pmatrix} \xrightarrow{\textcircled{2} + \textcircled{1}} \begin{pmatrix} A & B \\ A+C & B \end{pmatrix} \xrightarrow{\textcircled{1} - \textcircled{2}} \begin{pmatrix} -C & 0 \\ A+C & B \end{pmatrix}$$

$$\xrightarrow{\textcircled{1} \cdot (-1)} \begin{pmatrix} C & 0 \\ A+C & B \end{pmatrix} \xrightarrow{\textcircled{2} - \textcircled{1}(A+C)C^{-1}} \begin{pmatrix} C & 0 \\ 0 & B \end{pmatrix}$$

$$\begin{pmatrix} I & 0 \\ -(A+C)C^{-1} & I \end{pmatrix} \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} A & B \\ C & 0 \end{pmatrix} = \begin{pmatrix} C & 0 \\ 0 & B \end{pmatrix}$$

$$\left| \begin{pmatrix} I & 0 \\ -(A+C)C^{-1} & I \end{pmatrix} \right| |X| = |B| \cdot |C| \neq 0$$

$\therefore$ 只要 B 与 C 均可逆即可

(2) 设 $X^{-1} = \begin{pmatrix} P_1 & P_2 \\ P_3 & P_4 \end{pmatrix}$

$$\therefore XX^{-1} = \begin{pmatrix} AP_1 + BP_3 & AP_2 + BP_4 \\ CP_1 & CP_2 \end{pmatrix} \quad \therefore CP_1 = 0, \therefore P_1 = 0$$

$$\therefore XX^{-1} = \begin{pmatrix} BP_3 & AP_2 + BP_4 \\ 0 & CP_2 \end{pmatrix}$$

$$X^{-1}X = \begin{pmatrix} P_2C & 0 \\ P_3A + P_4C & P_3B \end{pmatrix} \quad \therefore P_2 = C^{-1}, P_3 = B^{-1}$$

$$\therefore B^{-1}A + P_4C = 0 \quad P_4C = -B^{-1}A$$

$$P_4 = -B^{-1}AC^{-1}$$

$$\therefore X^{-1} = \begin{pmatrix} 0 & C^{-1} \\ B^{-1} & -B^{-1}AC^{-1} \end{pmatrix}$$

$$7. (1) (AB - BA) \cdot (\varepsilon_1, \varepsilon_2, \varepsilon_3) = AB(\varepsilon_1, \varepsilon_2, \varepsilon_3) - BA(\varepsilon_1, \varepsilon_2, \varepsilon_3) \\ = (\varepsilon_1, \varepsilon_2, \varepsilon_3) \cdot (BA - AB)$$

$$BA = \begin{bmatrix}$$

(2)